

Literature Survey

Name: Kaustubh Kanodia

Registration Number: 23BIT0107

1. A Simulation-Based Real-Time Deep Reinforcement Learning Approach for Fighting Wildfires

This paper presents a digital twin-based wildfire response system that combines simulation with real-time deep reinforcement learning. It uses satellite imagery to support fire detection, then runs a discrete simulation to model fire spread, population impact, and firefighting response times before applying an A2C policy to improve decisions in changing conditions.

Strengths	Weaknesses
• The paper is strongly aligned with your project’s idea of using a digital twin as an active decision-making environment rather than just a passive simulator. • It explicitly considers time-sensitive disaster response, which matches your focus on real-time rescue coordination and dynamic hazards.	• The work appears focused on wildfire response rather than broader heterogeneous multi-agent disaster coordination across land, water, and air. • It does not appear to address large-scale cloud orchestration, fleet coordination, or hospital/resource balancing in the depth your system requires.

2. Digital Twin-Enabled Unmanned Aerial Vehicles Network for Disaster Management

This paper explores how digital twin-enabled UAV networks can improve disaster management by enhancing situational awareness, task offloading, and resource allocation. The key idea is to use a digital twin to support smarter UAV operation in difficult disaster environments, especially where communication and coordination are unstable.

Strengths	Weaknesses
• The paper is highly relevant to your aerial-agent layer because it treats UAVs as intelligent disaster response assets rather than just sensing devices. • Its emphasis on resource allocation and intelligent offloading fits well with your	• The focus is mainly on UAV networks, so it does not fully cover heterogeneous fleets such as ambulances, boats, or heavy lifters. • The paper seems more network/resource-optimization oriented than end-to-end rescue workflow oriented, so it may not

Strengths	Weaknesses
cloud coordinator concept and your goal of scalable deployment.	directly solve victim triage and transport problems.

3. Decentralized Intelligence: A Digital Twin-Assisted Hierarchical Federated Deep Reinforcement Learning Framework for Resource Optimization

This paper proposes a DT-assisted hierarchical federated deep reinforcement learning framework for optimizing UAV resource allocation in disaster response scenarios. It combines predictive analytics, hierarchical federated learning, and federated deep reinforcement learning to reduce communication overhead while improving utility and task execution.

Strengths	Weaknesses
• The hierarchical design is a strong match for your project because it reflects the same need for centralized orchestration with distributed agents. • The paper directly addresses scalability and communication overhead, which are major bottlenecks in multi-agent disaster systems.	• The framework is still centered on UAV resource optimization, so it may not generalize fully to rescue logistics, transport, and hospital routing. • Federated learning adds complexity that may be unnecessary if the core system is running a single coordinated cloud simulation pipeline.

4. A Novel Information System Framework for Natural Disaster Management

This paper proposes an information system framework for natural disaster management that integrates autonomous decision agents, pathway diversity, and digital twins. The experiments use real-world 2024 Brazil fire disaster data and simulate three management layers: operational, tactical, and strategic.

Strengths	Weaknesses
• This is especially relevant to your architecture because it explicitly separates decision-making into operational, tactical, and strategic layers,	• The paper appears to be a framework-oriented study, so it may not provide as much implementation detail for RL

Strengths	Weaknesses
which mirrors your backend, RL layer, and cloud coordinator structure. • The inclusion of autonomous decision agents makes it a good conceptual match for your multi-agent coordination system.	training, digital-twin simulation, or live control as your project needs. • Since it is based on a specific fire-disaster context, it may have limited coverage of floods, transport routing, and heterogeneous rescue fleets.

5. Digital Twin-based Decision Support Systems for Natural Disaster Management: A Systematic Review of Current Trends and Approaches

This systematic review surveys recent digital twin-based decision support systems for natural disaster management. It identifies disaster types, data collection methods, computational techniques, and decision-making tasks, while also highlighting current gaps and future directions in the field.

Strengths	Weaknesses
• This paper is useful because it gives you a broad research grounding and helps justify why your project is needed. • It can support your literature review by showing that digital twin-based disaster decision support is an emerging area with clear gaps in real-time operational coordination.	• As a review paper, it does not propose a concrete deployed multi-agent rescue system, so it is more useful for context than for direct implementation guidance. • It will not provide the same level of technical specificity as a systems or simulation paper for PPO, JAX, or cloud orchestration.

References

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